Actividad 3.3

trayectoria 1 – exponencial

① $x(t) = t$

$y(t) = e^{0.2t}$

② derivar

$\dot{x}(t) = 1$

$\dot{y}(t) = 0.2e^{0.2t}$

③ sustituir en $u(t)$

$u(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$

$u(t) = \sqrt{1 + (0.2e^{0.2t})^2}$

④ sustituir en angulo de orientación

$\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right)$

$\theta(\tan) = \tan^{-1}\left(0.2e^{0.2t}\right)$

para condiciones iniciales:

se sustituye 0 en las originales

$x(0) = t$

$= 0$

$y(0) = e^{(0.2)(0)}$

$= 1$

$\theta(0) = \tan^{-1}(0.2e^{0.2(0)})$

$\theta(0) = \tan^{-1}(0.2e^{0})$

$\theta(0) = \tan^{-1}(0.2)$

⑤ obtener expresion matematica para ω

$\omega(t) = \left(\frac{d\theta}{dt}\right)$

$\omega(t) = \frac{d(\tan^{-1}(0.2e^{0.2t}))}{dt}$

$d(0.2e^{0.2t}) = 0.04e^{0.2t}$

$$\omega = \frac{0.04e^{0.2t}}{1 + (0.2e^{0.2t})^2}$$

trayectoria cosenoidal 2 [8 a 17]

① $x(t) = t$

$y(t) = 3\cos[0.4(t-8)]$

② $\dot{x}(t) = 1$

$\dot{y}(t) = -1.2\sin(0.4(t-8))$

③ $u(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$

$u(t) = \sqrt{1 + (-1.2\sin(0.4(t-8)))^2}$

④ $\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right)$

$\theta(t) = \tan^{-1}\left(-1.2\sin(0.4(t-8))\right)$

⑤ $w(t) = \left(\frac{d\theta}{dt}\right)$

$$w(t) = \frac{d\left(\tan^{-1}\left(-1.2\sin(0.4(t-8))\right)\right)}{dt}$$

$d(-1.2\sin(0.4(t-8))$

$$w = \frac{-0.48\cos(0.4(t-8))}{1 + (-1.2\sin(0.4(t-8)))^2}$$

trayectoria 3 logarítmica

① $x(t) = t$

$y(t) = 2\ln(t-17)$

② $\dot{x}(t) = 1$

$\dot{y}(t) = \frac{2}{t-17}$

③ $u(t) = \sqrt{\dot{x}^2 + \dot{y}^2}$

$u(t) = \sqrt{1 + \left(\frac{2}{t-17}\right)^2}$

④ $\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right)$

$\theta(t) = \tan^{-1}\left(\frac{2}{t-17}\right)^2$

⑤ $w(t) = \frac{d\theta}{dt}$

$$w = \frac{-\frac{2}{(t-17)^2}}{1 + \left(\frac{2}{t-17}\right)^2}$$